

8.333 Recitation 2

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Moments from Cumulants

Recap: The n^{th} moment of a random variable X is $\langle X^n \rangle$. It is generated by the characteristic function or moment generating function,

$$\hat{p}(k) = \langle e^{ikX} \rangle = \int dx p_X(x) e^{-ikx},$$

$\downarrow$
if density p_X exists

which satisfies

$$\langle X^n \rangle = \frac{1}{(-i)^n} \frac{\partial^n \hat{p}}{\partial k^n} \Big|_{k=0}$$

The n^{th} cumulant is denoted $\langle X^n \rangle_c$ and is generated by the cumulant generating function:

$$\psi(k) = \log \hat{p}(k) = \log \langle e^{-ikx} \rangle,$$

with

$$\langle X^n \rangle_c \equiv \frac{1}{(-i)^n} \frac{\partial^n \psi}{\partial k^n} \Big|_{k=0}$$

There is a diagrammatic way of obtaining relations b/w moments and cumulants. We will show this trick, then explain it:

Represent the n^{th} cumulant as a connected cluster of n points. The n^{th} moment is the sum over all possible subdivisions of n points into smaller clusters. The contribution of each subdivision to the sum is the product of the cumulants that the clusters represent.

i.e. to get n^{th} moments, draw n points $\begin{smallmatrix} \bullet \\ \vdots \\ \bullet \end{smallmatrix}$

Sum over all possible groupings:

$$\langle X^5 \rangle = \text{diagram of 5 points in a circle} + 5 \text{diagram of 4 points in a circle and 1 point outside} + 10 \text{diagram of 2 points in a circle and 3 points in another circle} + \dots$$

A connected cluster of n points corresponds to the n^{th} cumulant

$$\text{diagram of 4 points in a circle} = \langle X^4 \rangle_c, \quad \text{diagram of 2 points in a circle} = \langle X^2 \rangle_c \langle X \rangle, \text{ etc...}$$

Examples

$$\langle X \rangle = \bullet$$

$$\langle X^2 \rangle = \text{two dots in a circle} + \bullet \bullet$$

$$\langle X^3 \rangle = \text{three dots in a triangle} + 3 \text{ two dots in a circle and one dot} + \bullet \bullet \bullet$$

$$\langle X^4 \rangle = \text{four dots in a square} + 4 \text{ three dots in a triangle and one dot} + \underbrace{\frac{4!}{2! 2! 2!}}_3 \text{ two dots in a circle and two dots} + \underbrace{\frac{4!}{2! 2!}}_6 \text{ two dots in a circle and two dots with lines} + \dots$$

These correspond to

$$\langle X \rangle = \langle X \rangle_c$$

$$\langle X^2 \rangle = \langle X^2 \rangle_c + \langle X \rangle_c^2$$

$$\langle X^3 \rangle = \langle X^3 \rangle_c + 3 \langle X^2 \rangle_c \langle X \rangle_c + \langle X \rangle_c^3$$

$$\langle X^4 \rangle = \langle X^4 \rangle_c + 4 \langle X^3 \rangle_c \langle X \rangle_c + 3 \langle X^2 \rangle_c^2 + 6 \langle X^2 \rangle_c \langle X \rangle_c^2 + \langle X \rangle_c^4$$

Why does this work? Write $\psi(k)$ in terms of its power series and exponentiate, and then match with the power series for $\tilde{p}$:

$$\begin{aligned} \tilde{p}(k) &= \exp(\psi(k)) \\ \sum_{m=0}^{\infty} \frac{(-ik)^m}{m!} \langle x^m \rangle &= \exp \left[\sum_{n=0}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle_c \right] \\ &= \prod_{n=0}^{\infty} \exp \left[\frac{(-ik)^n}{n!} \langle x^n \rangle_c \right] \\ &= \prod_{n=0}^{\infty} \sum_{p_n} \left[\frac{(-ik)^{n p_n}}{p_n!} \left(\frac{\langle x^n \rangle_c}{n!} \right)^{p_n} \right] \end{aligned}$$

Now we match powers of $(-ik)$:

$$\langle x^m \rangle = \sum_{p_n \in S} m! \prod_n \frac{1}{p_n! (n!)^{p_n}} \langle x^n \rangle_c^{p_n}$$

$$\text{where } S = \{ p_n \in \mathbb{N} \mid \sum n p_n = m \}.$$

You can convince yourself that the prefactor is the number of ways of breaking m points into $\{ p_n \}$ clusters of size n .

Multiple Random Variables

Sometimes we are interested not in a single observable $X: \Omega \rightarrow \mathbb{R}$, but rather a collection $(X_1, X_2, \dots, X_n)$, which we sometimes represent as a vector-valued RV $\vec{X}: \Omega \rightarrow \mathbb{R}^n$.

Two random variables X, Y are independent if and only if

$$\mathbb{P}[X \in A, Y \in B] = \mathbb{P}[X \in A] \mathbb{P}[Y \in B]$$

This denotes the event $(X^{-1}(A) \cap Y^{-1}(B)) \in \mathcal{F}$.

If X and Y are not independent, they are said to be correlated.

The rigorous theory of correlated RVs and conditional measures requires more preparation. Here we will discuss the elementary case of discrete random variables and motivate the extension to Lebesgue measurable collections of RVs.

1. Joint Probability Distributions

The Joint Cumulative Distribution Function is defined as

$$F(x_1, x_2, \dots, x_n) = \mathbb{P}[X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n]$$

For independent R.V.s, this factorizes into

$$F(x_1, \dots, x_n) = F_1(x_1) F_2(x_2) \dots F_n(x_n).$$

The Joint Probability Density is, for discrete Ω ,

$$p(x_1, x_2, \dots, x_n) = \mathbb{P}[X_1 = x_1, \dots, X_n = x_n].$$

For continuous Ω , it is defined as

$$p(x_1, \dots, x_n) = \frac{\partial^n F(x_1, \dots, x_n)}{\partial x_1 \partial x_2 \dots \partial x_n}, \quad \text{where it exists.}$$

You can check that this satisfies:

$$\mathbb{P}[A] = \int_{\vec{x} \in A} \prod_i dx_i p(x_1, \dots, x_n)$$

and can be interpreted

$$" (\prod_i dx_i) p(x_1, \dots, x_n) = \mathbb{P}[\vec{X} \in ([x_1, x_1+dx_1], \dots, [x_n, x_n+dx_n])] "$$

The joint characteristic function and joint moments and cumulants are defined as expected:

$$\hat{P}(\vec{k}) = \langle \exp(-i \sum_j k_j x_j) \rangle \quad \text{N-dim F.T. of joint density}$$

$$\langle x_1^{m_1} x_2^{m_2} \dots x_n^{m_n} \rangle = \left[\frac{\partial}{\partial(-ik_1)} \right]^{m_1} \dots \left[\frac{\partial}{\partial(-ik_n)} \right]^{m_n} \hat{P}(\vec{k}) \Big|_{k=0}.$$

$$\langle x_1^{m_1} x_2^{m_2} \dots x_n^{m_n} \rangle_c = \left[\frac{\partial}{\partial(-ik_1)} \right]^{m_1} \dots \left[\frac{\partial}{\partial(-ik_n)} \right]^{m_n} \log(\hat{P}(\vec{k})) \Big|_{k=0}.$$

2. Marginal and Conditional Measures

What if we are only interested in a subset of R.V.s?

For example, suppose we only need $(x_1, \dots, x_m)$ for $m < n$.

The joint CDF for this subset is clearly:

$$\begin{aligned} F_m(x_1, \dots, x_m) &\equiv \mathbb{P}[X_1 < x_1, \dots, X_m < x_m] \\ &= \mathbb{P}[X_1 < x_1, \dots, X_m < x_m, X_{m+1} < \infty, \dots, X_n < \infty] \\ &= F(x_1, \dots, x_m, \infty, \dots, \infty) \end{aligned}$$

The corresponding density is called the marginal probability density for the variables $X_1, \dots, X_m$. You can check that it is obtained by "integrating out" the variables $X_{m+1}, \dots, X_n$:

For discrete:
$$p(x_1, x_2, \dots, x_m) = \sum_{x_{m+1}} \sum_{x_{m+2}} \dots \sum_{x_n} p(x_1, \dots, x_n)$$

For continuous:
$$p(x_1, \dots, x_m) = \int \prod_{k=m+1}^n dx_k p(x_1, \dots, x_n)$$

For two independent R.V.s, the joint density is the product of the marginal densities:

$$P(x_1, x_2) = P_{X_1}(x_1) P_{X_2}(x_2).$$

For correlated R.V.s, this is not true, and we require the notion of conditional probability:

*
$$P[A|B] \equiv \frac{P[A \cap B]}{P[B]} \quad (\text{for } P[B] > 0)$$

This is the probability of event A given that B has occurred (i.e. conditional on B).

For discrete Ω , you can check at home that the conditional measure $P[\cdot|B]$ is a valid probability measure on the sample space. Thus for discrete R.V.s X_1 and X_2 , we can define a

conditional density

$$p(x_2|x_1) = P[X_2 = x_2 | X_1 = x_1] = \frac{P[X_2 = x_2, X_1 = x_1]}{P[X_1 = x_1]}$$

$$\Rightarrow p(x_2|x_1) = \frac{P(x_1, x_2)}{P(x_1)} \quad **$$

where $p(x_1, x_2)$ is the joint density. Now, I claim that we can make a similar definition in the continuous case, replacing discrete densities with continuous ones. This is subtle: if X_1 and X_2 are continuous, then $P(X_1 \in A | X_2 = x_2)$ appears ill-defined according to

equation ~~*~~, because $P[X_2 = x_2] = 0$ for continuous X_2 (that is, $X_2^{-1}(x_2)$ is a measure-zero set).

But we can condition on measure zero events by taking limits of finite measure events. Rewriting ~~(*)~~ in terms of integrals over densities:

$$P(X_2 \in A \mid X_1 \in B) = \frac{P(X_2 \in A \cap X_1 \in B)}{P(X_1 \in B)}$$

$$\int_A dx_2 P(x_2 \mid X_1 \in B) = \frac{\int_A \int_B dx_1 dx_2 P(x_1, x_2)}{\int_B dx_1 P(x_1)}$$

The above can be taken as a definition for the conditional density. If we take A, B to be narrow windows around x_2, x_1 , respectively, each of width Δx , we get

$$\Delta x P(x_2 \mid X_1 \in (x_1, x_1 + \Delta x)) \approx \frac{(\Delta x)^2 P(x_1, x_2)}{(\Delta x) P(x_1)}$$

This motivates ~~(**)~~. You can try to rigorize this argument at home.

A simple consequence of the definition ~~(*)~~ is Bayes' theorem:

$$P[A \mid B] = \frac{P[B \mid A] \cdot P[A]}{P[B]}$$

Expectations with respect to the conditional measure are called conditional expectations:

For X an RV, A an event

$$\begin{aligned} E[X|A] &\equiv \int X \, dP_{X|A} \\ &= \int dx \, x \, p(x|A) \end{aligned}$$

integration w.r.t
conditional measure

$E[X|A]$ is a number. We can pick $A = (Y=y)$, then $E[X|Y=y]$ is regarded as a function of $y \in \mathbb{R}$ into $\mathbb{R}$.

Sometimes we take a conditional expectation with respect to a random variable Y .

$$E[X|Y]$$

This is a real-valued function of the random variable Y , and is therefore itself a random variable. This obeys the law of iterated expectations, aka the tower rule:

$$E[E[X|Y]] = E[X]$$

"Proof":

$$\begin{aligned} E[E[X|Y]] &= \int E[X|Y] \, dP_Y \\ &= \int \int X \, dP_{X|Y} \, dP_Y \\ &= \iint x \, p(x|y) \, p(y) \, dx \, dy \\ &= \iint x \, p(x,y) \, dx \, dy \\ &= E[X]. \end{aligned}$$

Some Important Distributions

1. The Normal (Gaussian) Distribution

Then density of a Normal R.V. is a Gaussian function:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

The characteristic function of a Normal R.V. is also Gaussian:

$$\hat{p}(k) = \int \frac{dx}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} - ikx\right)$$

Sub $x \rightarrow x + \mu$:

$$= \frac{e^{-ik\mu}}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} dx e^{-\frac{x^2}{2\sigma^2} - ikx}$$

A good integral to memorize is:

$$\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2 + bx} = \sqrt{\frac{2\pi}{a}} e^{b^2/2a}$$

for $\text{Re } a > 0$

Thus

$$\hat{p}(k) = e^{-ik\mu} e^{-\sigma^2 k^2/2}$$

To get the cumulants, take the log

$$\psi(k) = \log \hat{p}(k) = -ik\mu - \frac{\sigma^2 k^2}{2}$$

We identify

$$\langle X \rangle_c = \mu \quad (\text{mean})$$

$$\langle X^2 \rangle_c = \sigma^2 \quad (\text{variance})$$

$$\langle X^{n \geq 3} \rangle_c = 0$$

All cumulants greater than two vanish - this is a unique property of Gaussian R.V.s. All moments can be expressed in terms of μ and σ .

If X is normally distributed with mean μ and variance σ^2 , we write

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

2. The multivariate Gaussian

A random vector $\vec{X} \in \mathbb{R}^n$ is said to be multivariate normal if every linear combination of its entries is Gaussian.

Such vectors have a joint density of the form:

$$p(\vec{x}) \equiv p(x_1, \dots, x_n) \equiv \frac{1}{\sqrt{(2\pi)^n \det C}} e^{-\frac{1}{2} (\vec{x} - \vec{\mu})^T C^{-1} (\vec{x} - \vec{\mu})}$$

where $\vec{\mu}$ is the mean vector:

$$\mu_i = \langle X_i \rangle$$

and C is the covariance matrix, a real symmetric matrix satisfying

$$C_{ij} \equiv \langle (X_i - \mu_i)(X_j - \mu_j) \rangle = \langle X_i X_j \rangle_c$$

The characteristic function is

$$\hat{p}(\vec{k}) = \exp\left(-i\vec{k}^T \vec{\mu} - \frac{1}{2} \vec{k}^T C \vec{k}\right)$$

This verifies that $C_{ij} = \langle X_i X_j \rangle_c$, and that all higher order cumulants vanish.

In the special case $\vec{\mu} = 0$, the distribution is symmetric (this is always true in the frame $\vec{y} = \vec{x} - \vec{\mu}$). In that case, all odd moments vanish. The rules before indicate that any even moment is obtained by summing over all ways of pairing the involved random variables:

$$\langle X_1 X_2 X_3 X_4 \rangle = C_{12} C_{34} + C_{13} C_{24} + C_{14} C_{23}$$

This is called "Wick's theorem" in the context of field theory.

3. The Binomial Distribution

Consider a biased coin which lands heads with probability p . Flip the coin N times. How many heads do you get?

Equivalently, consider N independent spins, each pointing up with probability

p. How many spins point up?

The random variables defined above follow a binomial distribution, which is a discrete distribution supported on $\{0, 1, 2, \dots\}$.

You can check that this has probability density

$$P[X=n] = p(n) = \binom{N}{n} p^n (1-p)^{N-n}$$

The characteristic function is

$$\begin{aligned}\hat{p}(k) &= \langle e^{-ikx} \rangle = \sum e^{-ikn} p(n) \\ &= \sum_{n=0}^{\infty} \binom{N}{n} p^n (1-p)^{N-n} e^{-ikn} \\ &= (pe^{-ik} + 1)^N\end{aligned}$$

Why? X is sum of N independent coin flips (Bernoulli RVs).

Let each flip correspond to an R.V. Y_i , with $Y_i = 1$ if i th flip lands heads and $Y_i = 0$ if tails.

Then

$$\begin{aligned}\hat{p}_x(k) &= \langle e^{-ikx} \rangle = \langle e^{-ik \sum Y_i} \rangle = \langle \prod_i e^{-ikY_i} \rangle \\ &= \prod_i \langle e^{-ikY_i} \rangle = \langle e^{-ikY_1} \rangle^N \\ &= [P[Y_1=1]e^{-ik \cdot 1} + P[Y_1=0]e^{-ik \cdot 0}]^N \\ &= (pe^{-ik} + 1-p)^N = (\hat{p}_y(k))^N\end{aligned}$$

The cumulant generating function is a sum over each coin flip:

$$\psi_x(k) = N \log(pe^{-ik} + 1-p) = N \psi_y(k).$$

The cumulants are:

$$\langle X \rangle = Np$$

$$\langle X^2 \rangle_c = Np(1-p)$$

As $N \rightarrow \infty$, $\frac{\sqrt{\langle X^2 \rangle_c}}{\langle X \rangle} \rightarrow 0 \Rightarrow X$ becomes sharply peaked.

Sums of R.V.s and the Central Limit Theorem (CLT)

Adding R.V.s

Consider two independent random variables X, Y .
What is the distribution of their sum $Z = X + Y$?
This is easy to answer using characteristic functions:

$$\begin{aligned}\hat{P}_Z(k) &= \langle e^{-ikZ} \rangle = \langle e^{-ikX} e^{-ikY} \rangle \stackrel{\text{independence}}{=} \langle e^{-ikX} \rangle \langle e^{-ikY} \rangle \\ &= \boxed{\hat{P}_X(k) \hat{P}_Y(k)}\end{aligned}$$

$\therefore$ Char. function of sum of indep. R.V.s is product of char. functions.

What does this tell us about the densities? Multiplication in Fourier space is convolution in real space:

$$\begin{aligned}P_Z(z) &= \int \frac{dk}{2\pi} \hat{P}_X(k) \hat{P}_Y(k) e^{ikz} = \int \frac{dk}{2\pi} \iint dx dy P_X(x) e^{-ikx} P_Y(y) e^{-iky} e^{ikz} \\ &= \frac{1}{2\pi} \iint dx dy P_X(x) P_Y(y) \int dk e^{ik(z-x-y)} \\ &= \iint dx dy P_X(x) P_Y(y) \delta(z-x-y)\end{aligned}$$

$$\boxed{P_Z(z) = \int dx P_X(x) P_Y(z-x)}$$

Example: Sum of two independent Gaussians $\Rightarrow$ Gaussian:

$$\begin{aligned}\text{Let } Z &= X + Y, & X &\sim \mathcal{N}(\mu_X, \sigma_X^2) \\ & & Y &\sim \mathcal{N}(\mu_Y, \sigma_Y^2)\end{aligned}$$

Then

$$\begin{aligned}\hat{P}_Z(k) &= \left(e^{-ik\mu_X} e^{-\sigma_X^2 k^2/2} \right) \left(e^{-ik\mu_Y} e^{-\sigma_Y^2 k^2/2} \right) \\ &= e^{-ik(\mu_X + \mu_Y)} e^{-(\sigma_X^2 + \sigma_Y^2) k^2/2}\end{aligned}$$

$$\therefore Z \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$$

Note: The Gaussian is the only distribution with finite moments that obeys this property (self similarity under convolution).

This motivates the CLT:

The CLT (will talk more about this later)

Consider a collection of N independent random variables $X_1, X_2, \dots, X_N$, with means μ_i and variances $\sigma_i^2 < \infty$.

What is the distribution of the sample average as $N \rightarrow \infty$?

$$\bar{X} \equiv \frac{1}{N} \sum_i X_i$$

The CLT states that this approaches a normal R.V. with mean $\frac{1}{N} \sum \mu_i$ and variance $\frac{1}{N} \sum \sigma_i^2$.

$$\bar{X} \stackrel{N \rightarrow \infty}{\sim} N\left(\frac{1}{N} \sum \mu_i, \frac{1}{N} \sum \sigma_i^2\right)$$

You will prove this on a future pset by showing that all cumulants of order ≥ 3 become vanishingly small compared to the first two. The stability of Gaussian R.V.s under addition gives some intuition.

The most common application is for independent, identically distributed R.V.s (iid):

$$\bar{X} \stackrel{N \rightarrow \infty}{\sim} N\left(\mu, \frac{\sigma^2}{N}\right)$$

The width of the distribution for large N vanishes as $\frac{\sigma}{\sqrt{N}}$

This is an instance of the Law of Large Numbers, which states that sample averages become sharply peaked around the true mean.

Note: There is a more general CLT which applies to weakly-dependent R.V.s, i.e. R.V.s whose correlations decay with N sufficiently fast. This version is harder to prove.